

Orally absorbed reactive glycation products (glycotoxins): An environmental risk factor in diabetic nephropathy

Endogenous advanced glycation endproducts (AGEs) include chemically crosslinking species (glycotoxins) that contribute to the vascular and renal complications of diabetes mellitus (DM). Renal excretion of the catabolic products of endogenous AGEs is impaired in patients with diabetic or nondiabetic kidney disease (KD). The aim of this study was to examine the oral absorption and renal clearance kinetics of food AGEs in DM with KD and whether circulating diet-derived AGEs contain active glycotoxins. Thirty-eight diabetics (DM) with or without KD and five healthy subjects (NL) received a single meal of egg white (56 g protein), cooked with (AGE-diet) or without fructose (100 g) (CL-diet). Serum and urine samples, collected for 48 hr, were monitored for AGE immunoreactivity by ELISA and for AGE-specific crosslinking reactivity, based on complex formation with ¹²⁵I-labeled fibronectin. The AGE-diet, but not the CL-diet, produced distinct elevations in serum AGE levels in direct proportion to amount ingested ($r = 0.8$, $P < 0.05$); the area under the curve for serum ($\approx 10\%$ of ingested AGE) correlated directly with severity of KD; renal excretion of dietary AGE, although normally incomplete (only $\approx 30\%$ of amount absorbed), in DM it correlated inversely with degree of albuminuria, and directly with creatinine clearance ($r = 0.8$, $P < 0.05$), reduced to $< 5\%$ in DM with renal failure. Post-AGE-meal serum exhibited increased AGE-crosslinking activity (two times above baseline serum AGE, three times above negative control), which was inhibited by aminoguanidine. In conclusion, (i) the renal excretion of orally absorbed AGEs is markedly suppressed in diabetic nephropathy patients, (ii) daily influx of dietary AGEs includes glycotoxins that may constitute an added chronic risk for renal-vascular injury in DM, and (iii) dietary restriction of AGE food intake may greatly reduce the burden of AGEs in diabetic patients and possibly improve prognosis.